Company Employee Facilities Utilization & Cost Analysis

Project Documentation | Power BI Dashboard

This document covers the Company Employee Facilities Utilization & Cost Analysis project: dataset format, data cleaning, the cleaned datasets, the data model, the report layout and the final dashboard snapshots. The solution helps Company see how employees use the seven company facilities, who pays for that use, and whether spending stays within the monthly budget. The data is synthetic and covers January to December 2026.

Item	Detail
Tool	Power BI Desktop (Power Query for cleaning, DAX for the Calendar table and measures)
Source files	Employees (CSV), Facilities (CSV), FacilityUsage (CSV)
Model tables	Employees, Facilities, FacilityUsage, Calendar
Contents	1 Dataset Format | 2 Data Cleaning | 3 New Dataset After Cleaning | 4 Data Model & Relationships | 5 DAX Measures | 6 Report Layout | 7 Final Version Snapshot

1. Dataset Format

Three source datasets were loaded from CSV files. A fourth table, the Calendar table, was created inside Power BI. EmployeeID, FacilityID and Month are the linking fields.

1.1 Employees

Organisational information of each employee (one row per employee).

Field	Data Type	Description
EmployeeID	Text	Unique identifier of each employee (key)
EmployeeName	Text	Full name of the employee
Department	Text	Department in which the employee works (10 departments)
JobRole	Text	Current job role
OfficeLocation	Text	Delhi, Gurugram, Noida, Mumbai, Bengaluru, Hyderabad, Pune or Chennai
EmploymentType	Text	Full-Time, Part-Time, Contract or Intern
JoiningDate	Date	Date the employee joined Company
EmployeeStatus	Text	Active, Inactive or On Leave

1.2 Facilities

The seven facilities Company provides (one row per facility).

Field	Data Type	Description
FacilityID	Text	Unique identifier of each facility (key)
FacilityName	Text	Transport, Cafeteria, Parking, Gym & Fitness Center, Medical Center, Arcade Room, Swimming Pool
Category	Text	Transport, Food & Beverage, Parking, Fitness, Medical or Recreation
Provider	Text	Company that runs the facility
MonthlyOperatingCost	Decimal (INR)	Flat monthly cost of running the facility
MonthlyBudget	Decimal (INR)	Flat monthly budget for the facility
FacilityStatus	Text	Active or Inactive

1.3 FacilityUsage

Facility usage per employee, facility and month (one row per employee + facility + month).

Field	Data Type	Description
UsageID	Text	Unique identifier of each usage record (key)
EmployeeID	Text	Links the usage to an employee (foreign key)
FacilityID	Text	Links the usage to a facility (foreign key)
Month	Date	First day of the month, within 2026
UsageFrequency	Whole Number	Number of times the facility was used in the month (1 or more)
CostPerUse	Decimal (INR)	Cost of one use (greater than 0)
PaidBy	Text	Company or Employee

1.4 Calendar Table (created in Power BI)

A single Calendar table, created using DAX, provides the common time dimension and the month count used to scale monthly costs. Fields: Date, Year, Month Number, Month Name, Month Year, Is Month Start.

2. Data Cleaning - What We Have Done

All three CSV files were imported using Get Data > Text/CSV > Transform Data, with every column loaded as Text first so nothing was silently converted or dropped. The raw files were never edited. Each table has an untouched raw query and a clean query built from it, and all fixes were made in Power Query before the model was built.

2.1 Employees

- **Structure:** Promoted headers, applied data types (JoiningDate as Date) and removed duplicate EmployeeID rows (E034 and E039 were repeated), taking the table from 152 to 150 rows.
- **Text cleaning:** Trim and collapse spaces removed leading, trailing and double spaces in names.
- **Standardising:** Lookup tables fixed Department, EmploymentType and EmployeeStatus, e.g. "it" and "Information Technology" to IT, "Full Time" and "Permanent" to Full-Time, "active" to Active.

- **Locations:** Bangalore to Bengaluru, Gurgaon to Gurugram, New Delhi to Delhi. Blank locations were set to "Unknown".
- **Dates:** Text dates such as 15/07/2021 were converted. A placeholder of 1900-01-01 and a future date of 2027-03-15 were set to null.

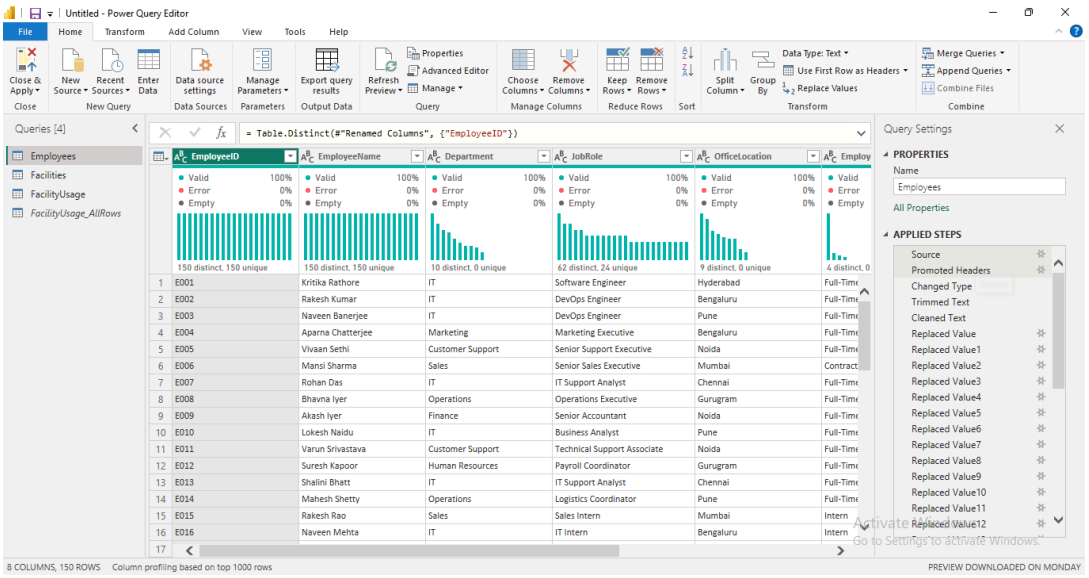


Figure 1: Employees query - applied steps in Power Query

2.2 Facilities

- **Standardising:** Category, Provider and FacilityStatus were trimmed and mapped to consistent spellings (e.g. "food & beverage" to Food & Beverage, "active" to Active).
- **Numbers:** Text amounts such as "₹1,80,000" and "Rs 2,20,000" were converted to numbers. The negative Gym operating cost (-140000) was corrected to a positive value.
- **Duplicates:** The duplicate Cafeteria row (F002 appeared twice) was removed, taking the table from 8 to 7 rows.
- **Missing values:** The Swimming Pool operating cost and the Arcade Room budget were kept as null, not invented. A DataNote column records what is missing.

2.3 FacilityUsage

Area	Cleaning performed
Duplicates	10 exact duplicate rows removed; 6 rows repeating the same employee + facility + month under a new UsageID removed.
PaidBy	82 inconsistent values ("company", "COMPANY", "Company ", "employee", "emp") standardised to Company or Employee.
Month	46 mixed formats (01-03-2026, 2026/04/01, 1-Jun-2026, Sep-2026) converted to a date using locale settings.
Rejected keys	20 rows with a missing or unknown EmployeeID and 8 rows with an unknown FacilityID (F008, F010) rejected.
Rejected values	11 rows with impossible, non-first-of-month or non-2026 Month, and 23 rows with zero or negative frequency or cost rejected.
Unusual values	Very high values (e.g. 44 transport trips, 61 cafeteria visits, a ₹8,000 medical visit) are kept and flagged, not deleted.

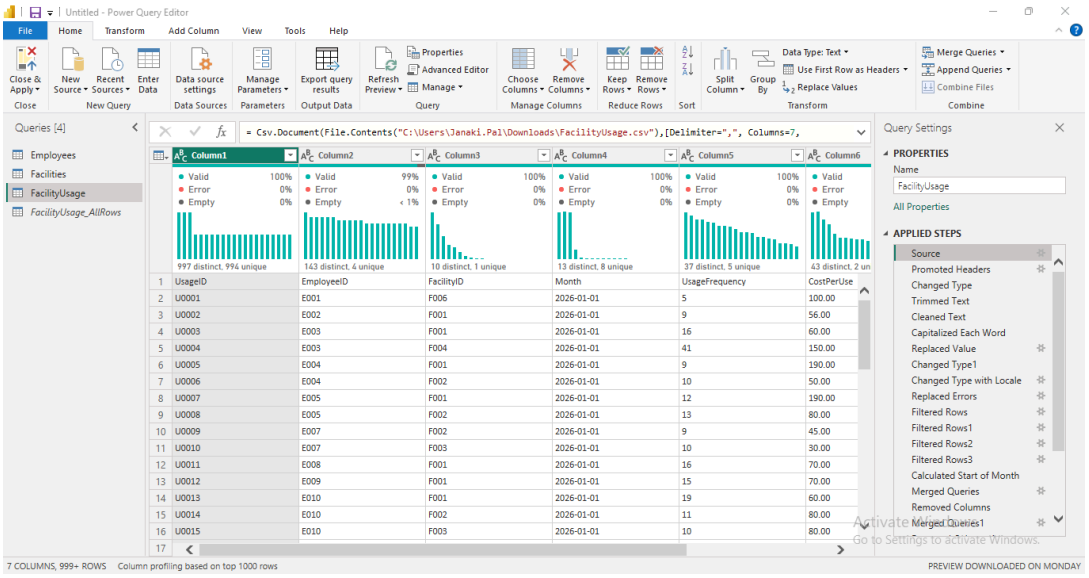


Figure 2: FacilityUsage query - cleaning steps in Power Query

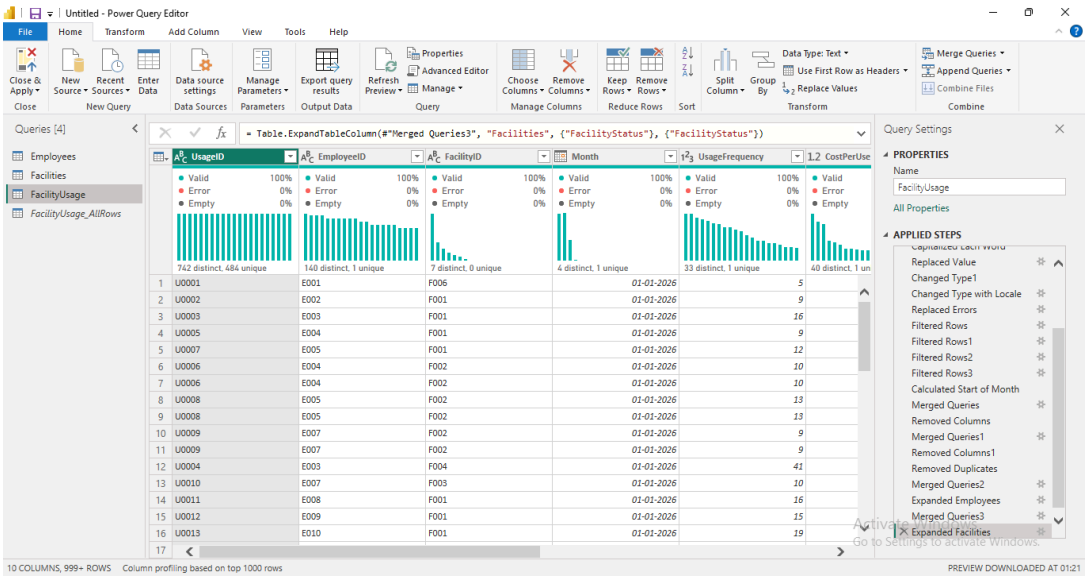


Figure 3: FacilityUsage query - final merge and expand steps

2.4 Columns added and quality checks

- **ReviewFlag:** added to FacilityUsage for rows that look unusual but may be valid: high frequency, usage by an Inactive employee, usage of an Inactive facility, and usage before the joining date.
- **DataNote:** added to Facilities to show where an operating cost or budget is missing.
- **FacilityUsage_Rejected:** an audit query holding the 68 removed rows with a reason. It is not loaded into the model.
- **Quality checks:** keys verified as unique, every usage row verified against an existing employee and facility, and categorical values verified as standardised.
- **Rule:** reject, don't guess (untrusted keys or months go to the audit query); flag, don't delete (unusual but possible values stay with a ReviewFlag).

3. New Dataset After Cleaning

After cleaning, the model contains four analysis-ready tables.

Table	Role	State after cleaning
Employees	Dimension (1 side)	150 unique employees; consistent Department, EmploymentType, EmployeeStatus and OfficeLocation. Status: Active 132, Inactive 13, On Leave 5.
Facilities	Dimension (1 side)	7 unique facilities; cost and budget as decimal numbers; DataNote added; missing values preserved as null.
FacilityUsage	Fact (many side)	3,774 rows (from 3,852); one row per employee + facility + month; ReviewFlag added (347 rows carry a flag).
Calendar	Dimension (1 side)	Created in DAX and marked as date table; Month Name and Month Year sorted by Month Number.

Table	Raw rows	Clean rows	Change
Employees	152	150	-2 duplicates
Facilities	8	7	-1 duplicate
FacilityUsage	3,852	3,774	-78 (10 + 62 + 6)

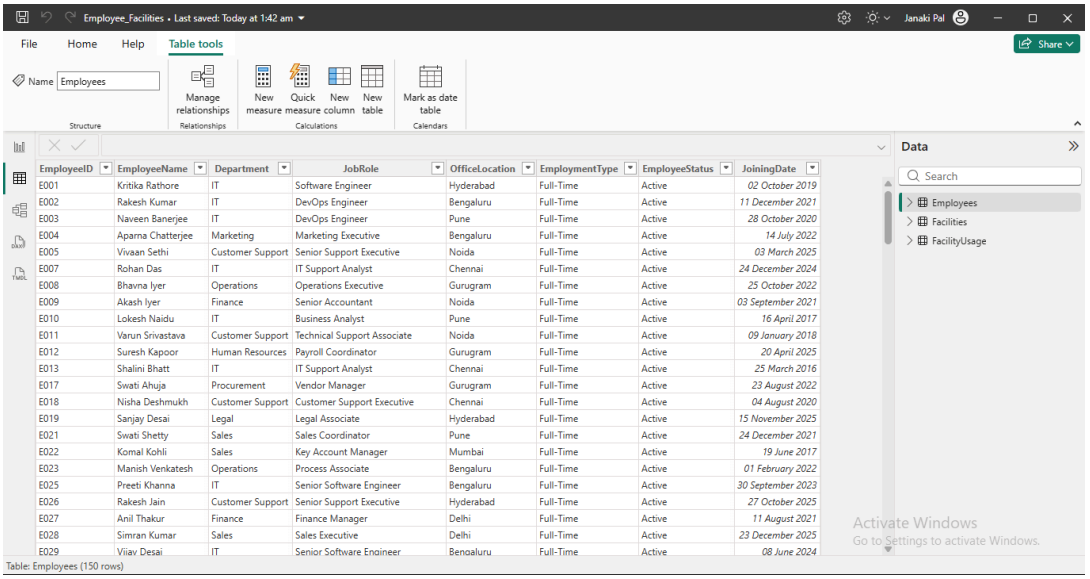


Figure 4: Employees table after cleaning (150 rows)

4. Data Model Building & Relationships

The final model follows a simple star schema: the FacilityUsage fact table sits in the centre, filtered by the Employees, Facilities and Calendar dimensions.

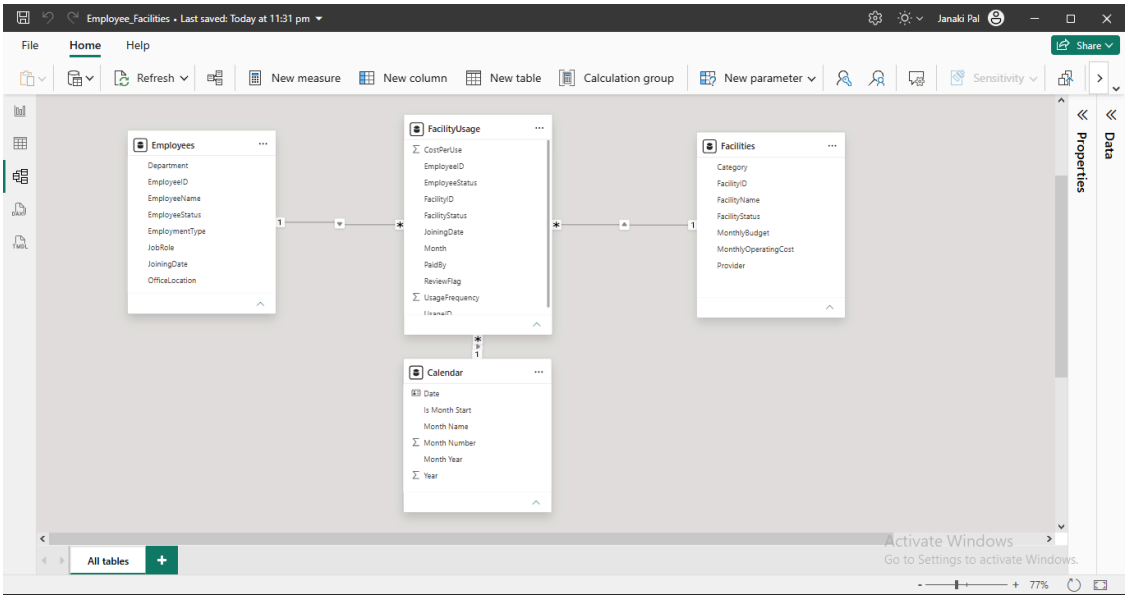


Figure 5: Model view

Relationship (From to To)	Cardinality	Filter	Status / Purpose
Employees[EmployeeID] to FacilityUsage[EmployeeID]	One-to-Many	Single	Active. Employee details filter their usage.
Facilities[FacilityID] to FacilityUsage[FacilityID]	One-to-Many	Single	Active. Facility details filter their usage.
Calendar[Date] to FacilityUsage[Month]	One-to-Many	Single	Active. Time-based analysis by month.

Limitation: MonthlyOperatingCost and MonthlyBudget belong to the facility, so a Department or Location slicer filters usage cost but not operating cost or budget. For this reason the Cost & Budget page has no Department or Location slicers.

5. DAX Measures

All measures are held in an empty _Measures table. Usage Cost is calculated, never stored. Because operating cost and budget are flat monthly amounts, they are multiplied by the number of months in the period.

Measure	Definition
Total Uses	SUM of UsageFrequency
Usage Cost	SUMX of UsageFrequency x CostPerUse
Company-Paid / Employee-Paid Usage Cost	Usage Cost filtered by PaidBy = Company or Employee
Active Users	DISTINCTCOUNT of EmployeeID in FacilityUsage

Measure	Definition
Avg Uses per User	Total Uses divided by Active Users
Months in Period	Count of Calendar rows where Is Month Start is true
Total Operating Cost	Sum of MonthlyOperatingCost x Months in Period
Total Company Spend	Total Operating Cost + Company-Paid Usage Cost
Comparable Budget / Spend	Budget and spend for facilities that have both an operating cost and a budget (excludes Arcade Room and Swimming Pool)
Budget Variance	Comparable Budget - Comparable Spend
Budget Utilization %	Comparable Spend divided by Comparable Budget
Budget Status	Over budget above 100%, Near budget 95-100%, Within budget below 95%, Data missing if blank
Flagged Rows	Count of FacilityUsage rows where ReviewFlag is not blank

Expected values for 2026 with no slicers: Total Uses 42,923; Usage Cost ₹41,11,960 (company ₹21,51,974, employee ₹19,59,986); Active Users 149; Average Uses per User 288.1; Comparable Budget ₹2,11,80,000; Budget Utilization 98.2%.

6. Report Layout

The report has four pages. A page navigator sits in the header of every page, slicers sit at the top-left, and the charts are arranged in a grid below KPI cards. Dark blue is used for Company, orange for Employee, and green, amber and red only for budget status.

Page 1 - Overview

Visual	Type	Fields / Purpose
Month Year, Department, Office Location, Facility	Slicers	Filter the whole page
Total Uses, Usage Cost, Company-Paid, Employee-Paid, Active Users	Cards	Headline KPIs
Usage by facility	Bar chart	Total Uses by FacilityName
Company vs Employee Paid	Clustered column	Company-paid and employee-paid usage cost by month
Monthly Usage Cost	Donut chart	Usage cost split by PaidBy

Page 2 - Utilization

Visual	Type	Fields / Purpose
Usage By Department	Matrix	Uses by Department x Facility with colour scale
Top 10 Employees	Table	Employees with the most uses and their usage cost
Avg Uses per User by OfficeLocation	Bar chart	Compares average use across the eight locations

Visual	Type	Fields / Purpose
Avg Uses per User by EmploymentType	Bar chart	Compares Full Time, Contract, Part Time and Intern

Page 3 - Cost & Budget

Visual	Type	Fields / Purpose
Month Year, FacilityName	Slicers	Filter the page (no Department or Location slicers)
Comparable Budget, Budget Variance, Total Company Spend, Budget Utilization %	Cards	Budget KPIs
Budget Utilization % by Month Name	Line chart	Utilization against the 100% line
Budget vs Spend	Clustered column	Comparable Spend and Comparable Budget by month
Facility Budget	Table	Operating cost, budget, variance, utilization %, Budget Status and months over budget, with status colours

Page 4 - Data Review

Visual	Type	Fields / Purpose
Flagged Rows, Usage Records	Cards	Count of flagged rows and total usage rows
Flagged for review	Table	UsageID, EmployeeID, FacilityID, Month, cost and ReviewFlag. A note states that rejected records are excluded from the report calculations

7. Final Version Snapshot

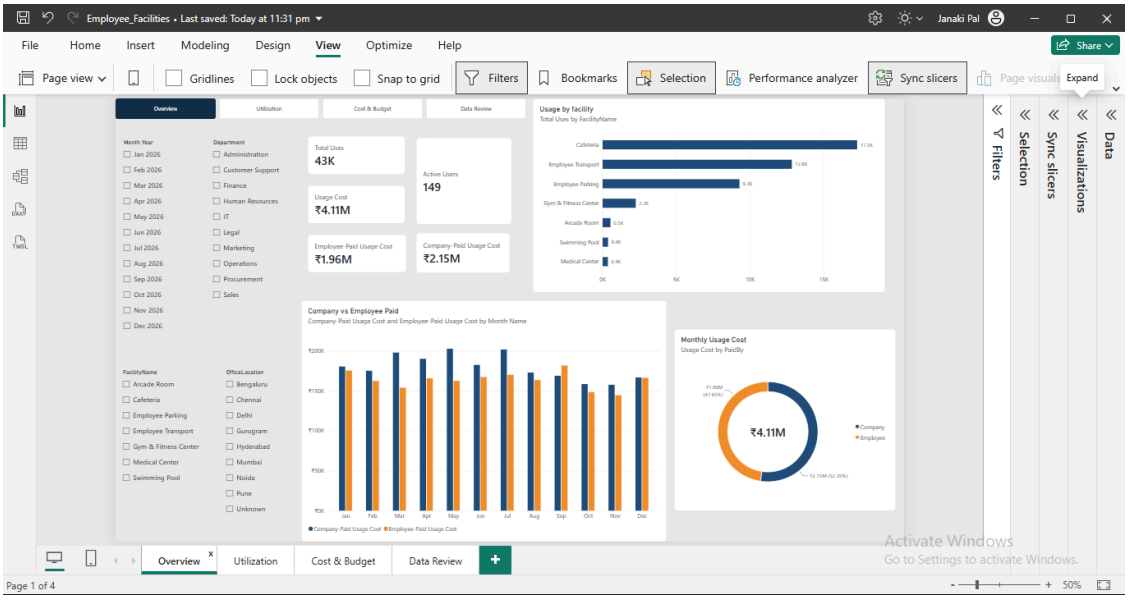


Figure 6: Dashboard Page 1 - Overview

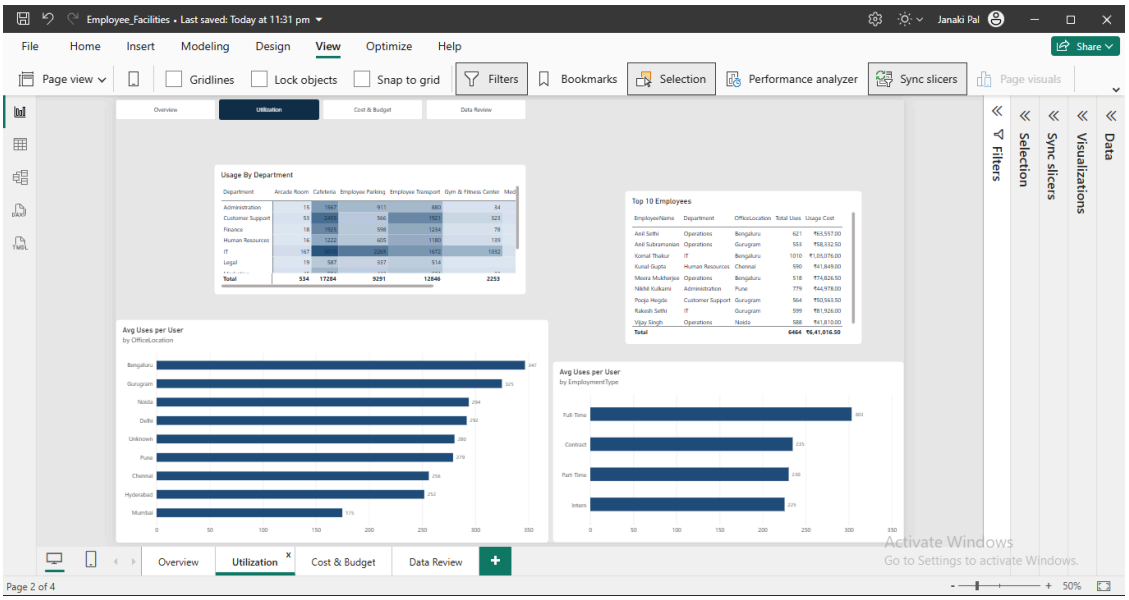


Figure 7: Dashboard Page 2 - Utilization

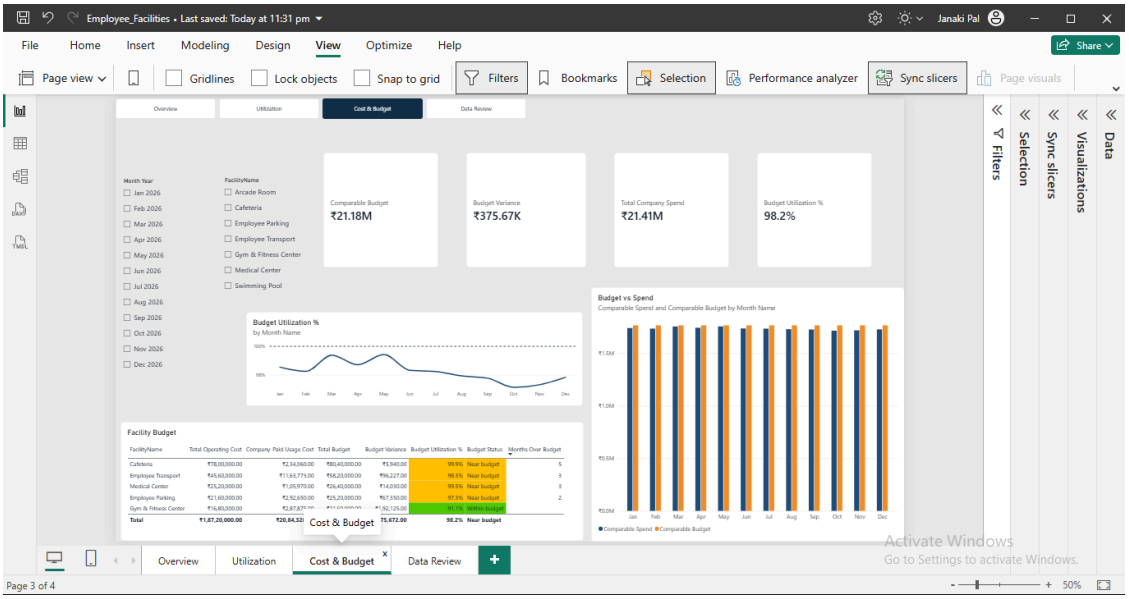


Figure 8: Dashboard Page 3 - Cost & Budget

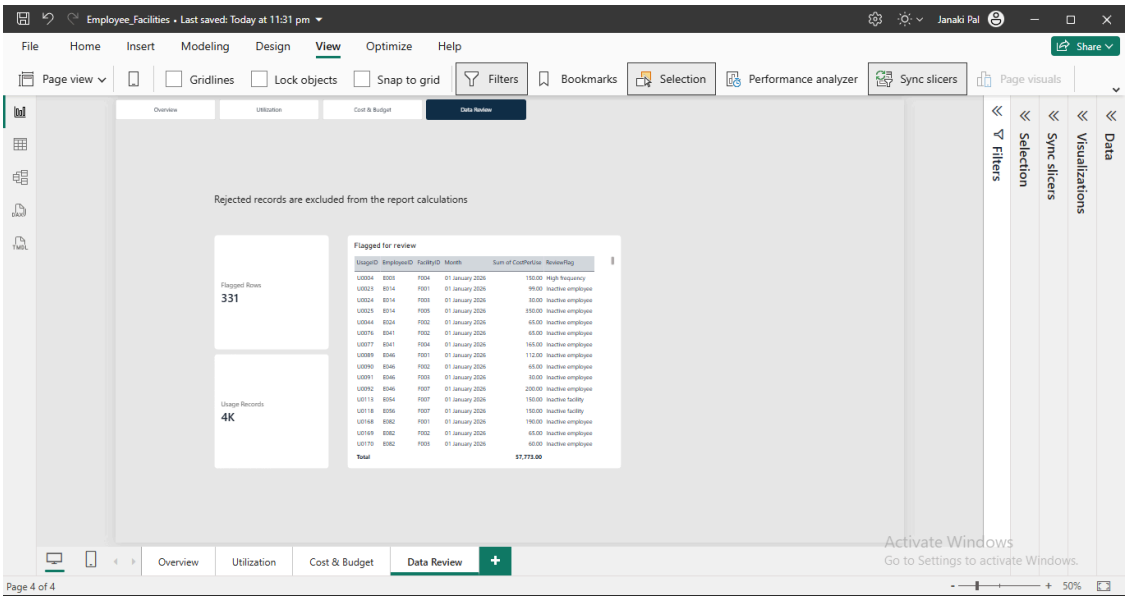


Figure 9: Dashboard Page 4 - Data Review

Findings

- Cafeteria and transport account for about 70% of all facility use; medical, arcade and pool together are about 3%.
- Company and employees split usage cost about 52% / 48%. Transport and medical are mostly company-paid; the cafeteria is mostly employee-paid.
- Spending sits at 91-100% of budget over the year. The Cafeteria was over budget in 5 months, Transport and Medical in 3, Parking in 2 and the Gym in none.
- Employee E125 has 1,010 uses, against an average of 288. The data is valid, so it is kept and called out.

Open items

- Confirm the Swimming Pool operating cost and the Arcade Room budget; budget checks for these two facilities are not possible until then.
- Review usage by Inactive employees and the Inactive Swimming Pool, and usage by E050 before the joining date. These need a business answer, not a data fix.

Outcome

The cleaned model supports analysis of facility usage by department, location and employment type, monthly usage trends, company-paid versus employee-paid cost, facility operating cost, and total spending against budget, giving Company a clear view of how its employee facilities are used and paid for.